



INTERFACING AT 100 GIGABITS PER SECOND

Almost every hardware interface standard is rushing towards a 100Gbps world—be it memory, storage, input/output, Ethernet, wireless or multimedia. So you can start looking forward to the day when it would take just a second to transfer a high-definition movie to your hard drive!

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Jef Raskin of Macintosh fame is known to have remarked that “As far as the customer is concerned, the interface is the product.” How true! After all, the brainiest man is only as useful as his ability to communicate his ideas to others and coordinate with them to implement these.

In the computing world, these abilities are realised, to a large extent, through interfaces. “An interface, simply put, is a point of interaction or communication between components or devices that works via an input/

output system and an associated protocol. The interface is central to the workability/effectiveness/utility of the device. Seamless (fast and high-fidelity) interaction is the ultimate objective,” defines Rajesh Gupta, director-Sales & Marketing Group, Intel South Asia.

While the word ‘interface’ is most likely to pop up images of snazzy user interfaces, colourful dialogue boxes, and easily navigable menus, there is another fundamental type of interface at work—the hardware interface. Does the mention of universal serial bus (USB) ring a bell? That is just one of the many types of hardware interfaces that work

at levels ranging from component to device, enabling these to communicate and coordinate with other components and devices in an *ad-hoc* or plug-and-play fashion. The capabilities of many a device, including memory, input/output, communication, audio/video and others, is largely dependent on its interfaces.

An interface is characterised by features like speed, quality of transmission, compatibility, adherence to standards and so on. While standardisation and plug-and-play capability are indispensable aspects of an interface, ‘speed’ is turning out to be a critical